CLOUD COMPUTING LABORATORY

ASSESSMENT TASK – EXPERIMENTS

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EXPERIMENT 1

Cloud-Based Library Book Reservation System

AIM

To create and deploy a simple cloud software application for a Library Book Reservation System using a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model.

OBJECTIVE

To develop a cloud-based application that allows users to view and manage library book details and check the reservation status of books.

SOFTWARE REQUIREMENTS

- Cloud Service Provider
- Web browser
- HTML
- CSS
- JavaScript
- Cloud database/service

REQUIRED FIELDS

The application contains the following fields:

Field	**Description**
Book Title	Name of the book
Author	Name of the author
Publication	Publisher name
Year	Year of publication

Edition	Edition of the book
No. of Copies	Number of available copies
Rack No.	Location of the book
Status	Taken / Returned

PROCEDURE

1. Select a suitable Cloud Service Provider.
2. Create a cloud-based application project.
3. Design a user interface for entering library book information.
4. Create fields for book title, author, publication, year, edition, number of copies, rack number and status.
5. Connect the application with a cloud database or cloud storage service.
6. Add sample book records to the application.
7. Implement the reservation functionality.
8. Update the status of a book as **Taken** when it is reserved.
9. Update the status as **Returned** when the book is returned.
10. Deploy the application on the cloud.
11. Access the application through a web browser.

SAMPLE DATA

Book Title	Author	Publication	Year	Edition	Copies	Rack No.	Status
Cloud Computing	Rajkumar Buyya	Wiley	2023	2nd	5	A-12	Available
Computer Networks	Andrew S. Tanenbaum	Pearson	2022	6th	3	B-05	Taken
Database Systems	Ramez Elmasri	Pearson	2021	7th	4	C-08	Returned

EXPECTED OUTPUT

The deployed cloud application displays the library book details and allows the user to reserve and return books. The book status is updated accordingly.

RESULT

Thus, a cloud-based Library Book Reservation System was successfully created and deployed using a Cloud Service Provider, demonstrating the SaaS model.

EXPERIMENT 2

Cloud-Based Payroll Processing System

AIM

To create and deploy a simple cloud software application for a Payroll Processing System using a Cloud Service Provider and demonstrate the Software as a Service (SaaS) model.

OBJECTIVE

To develop a cloud-based application for storing employee details and calculating gross pay and net pay.

SOFTWARE REQUIREMENTS

- Cloud Service Provider
- Web browser
- HTML
- CSS
- JavaScript
- Cloud database/service

REQUIRED FIELDS

Field	**Description**
Employee Name	Name of the employee
Experience in Present Organization	Current organization experience
Overall Experience	Total work experience
Basic Pay	Basic salary
HRA	House Rent Allowance
DA	Dearness Allowance

TA	Travel Allowance
Gross Pay	Total salary before deductions
Net Pay	Salary after deductions

FORMULAS

$$Gross\ Pay = Basic\ Pay + HRA + DA + TA$$

$$Net\ Pay = Gross\ Pay - Total\ Deductions$$

PROCEDURE

1. Select a suitable Cloud Service Provider.
2. Create a cloud application project.
3. Design a payroll input form.
4. Add fields for employee name, experience and salary details.
5. Enter the basic pay and allowances.
6. Implement the calculation of gross pay.
7. Add the required deductions.
8. Calculate the net pay.
9. Store the employee payroll details in the cloud.
10. Deploy the application.
11. Access the payroll application through a web browser.

SAMPLE DATA

Employee Name	Present Org. Experience	Overall Experience	Basic Pay	HRA	DA	TA	Gross Pay	Net Pay
Arun Kumar	3 Years	5 Years	₹30,000	₹6,000	₹3,000	₹2,000	₹41,000	₹39,000
Priya S	2 Years	4 Years	₹28,000	₹5,600	₹2,800	₹1,500	₹37,900	₹36,400

EXPECTED OUTPUT

The cloud application accepts employee and salary details and automatically calculates the gross pay and net pay.

RESULT

Thus, a cloud-based Payroll Processing System was successfully created and deployed using a Cloud Service Provider, demonstrating the SaaS model.

EXPERIMENT 3

Demonstration of Virtualization Using a Type-2 Hypervisor

AIM

To demonstrate virtualization by installing a Type-2 Hypervisor and creating a virtual machine with 1 CPU, 2 GB RAM and 15 GB storage.

OBJECTIVE

To understand the working of a Type-2 Hypervisor and create a virtual machine with the specified hardware resources.

SOFTWARE REQUIREMENTS

- Host Operating System: Windows/Linux
- Oracle VirtualBox or VMware Workstation
- Guest Operating System ISO image
- Minimum required RAM and storage

CONFIGURATION

Parameter	**Configuration**
Hypervisor	Oracle VirtualBox
Hypervisor Type	Type-2
CPU	1
RAM	2 GB
Storage	15 GB
Guest OS	Windows/Linux

PROCEDURE

1. Download and install Oracle VirtualBox on the host operating system.
2. Open VirtualBox.

3. Click **New** to create a virtual machine.
4. Enter a suitable name for the VM.
5. Select the required guest operating system.
6. Allocate **1 CPU** to the virtual machine.
7. Allocate **2 GB RAM**.
8. Create a virtual hard disk with **15 GB storage**.
9. Attach the operating system ISO image.
10. Start the virtual machine.
11. Install the guest operating system.
12. After installation, start the VM and verify that the guest operating system is running.

VIRTUALIZATION ARCHITECTURE

Host Operating System
↓
Type-2 Hypervisor (VirtualBox)
↓
Virtual Machine
↓
Guest Operating System

EXPECTED OUTPUT

The virtual machine starts successfully with:

- 1 CPU
- 2 GB RAM
- 15 GB storage

RESULT

Thus, virtualization was successfully demonstrated by installing a Type-2 Hypervisor and creating a virtual machine with 1 CPU, 2 GB RAM and 15 GB storage.

EXPERIMENT 4

Cloning of a Virtual Machine

AIM

To demonstrate virtualization by creating a clone of an existing virtual machine using a Type-2 Hypervisor and testing the cloned VM.

OBJECTIVE

To understand VM cloning and verify that the cloned virtual machine can be loaded and executed independently.

SOFTWARE REQUIREMENTS

- Oracle VirtualBox or VMware Workstation
- Existing Virtual Machine
- Host Operating System
- Guest Operating System

PROCEDURE

1. Open Oracle VirtualBox.
2. Identify the virtual machine created in Experiment 3.
3. Shut down the virtual machine completely.
4. Select the virtual machine.
5. Right-click the VM and select **Clone**.
6. Enter a new name for the cloned virtual machine.
7. Select **Full Clone** to create an independent copy.
8. Complete the cloning process.
9. Start the cloned virtual machine.
10. Verify that the guest operating system loads successfully.
11. Check the system configuration and applications.
12. Test the cloned VM to confirm that it works independently from the original VM.

CLONING PROCESS

Original Virtual Machine
↓
Select Clone
↓
Create Full Clone
↓
New/Cloned Virtual Machine
↓
Start Cloned VM
↓
Test and Verify

EXPECTED OUTPUT

The cloned virtual machine starts successfully and loads the same guest operating system and configuration as the original VM.

RESULT

Thus, a virtual machine was successfully cloned using a Type-2 Hypervisor, and the cloned VM was tested successfully.